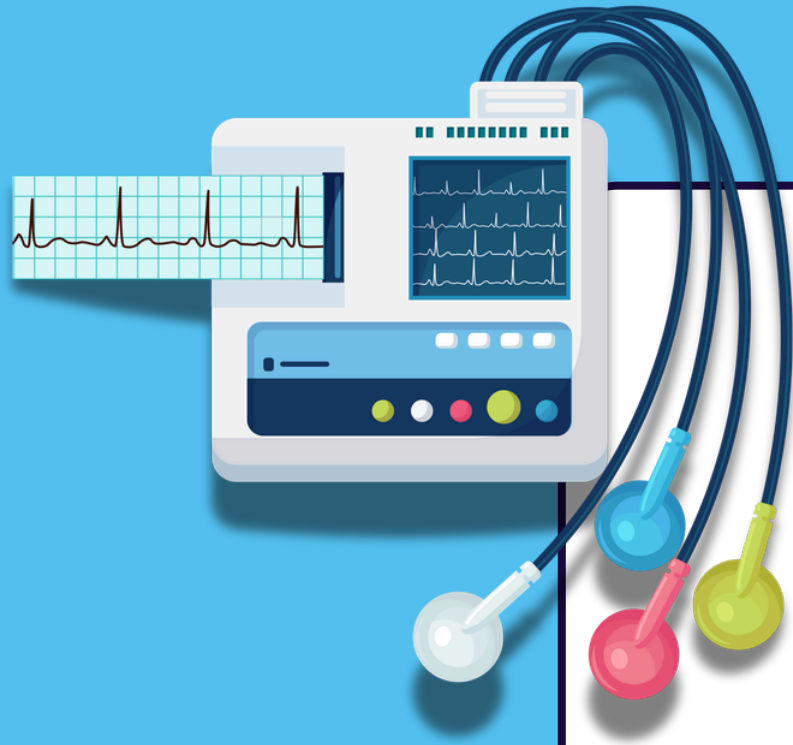




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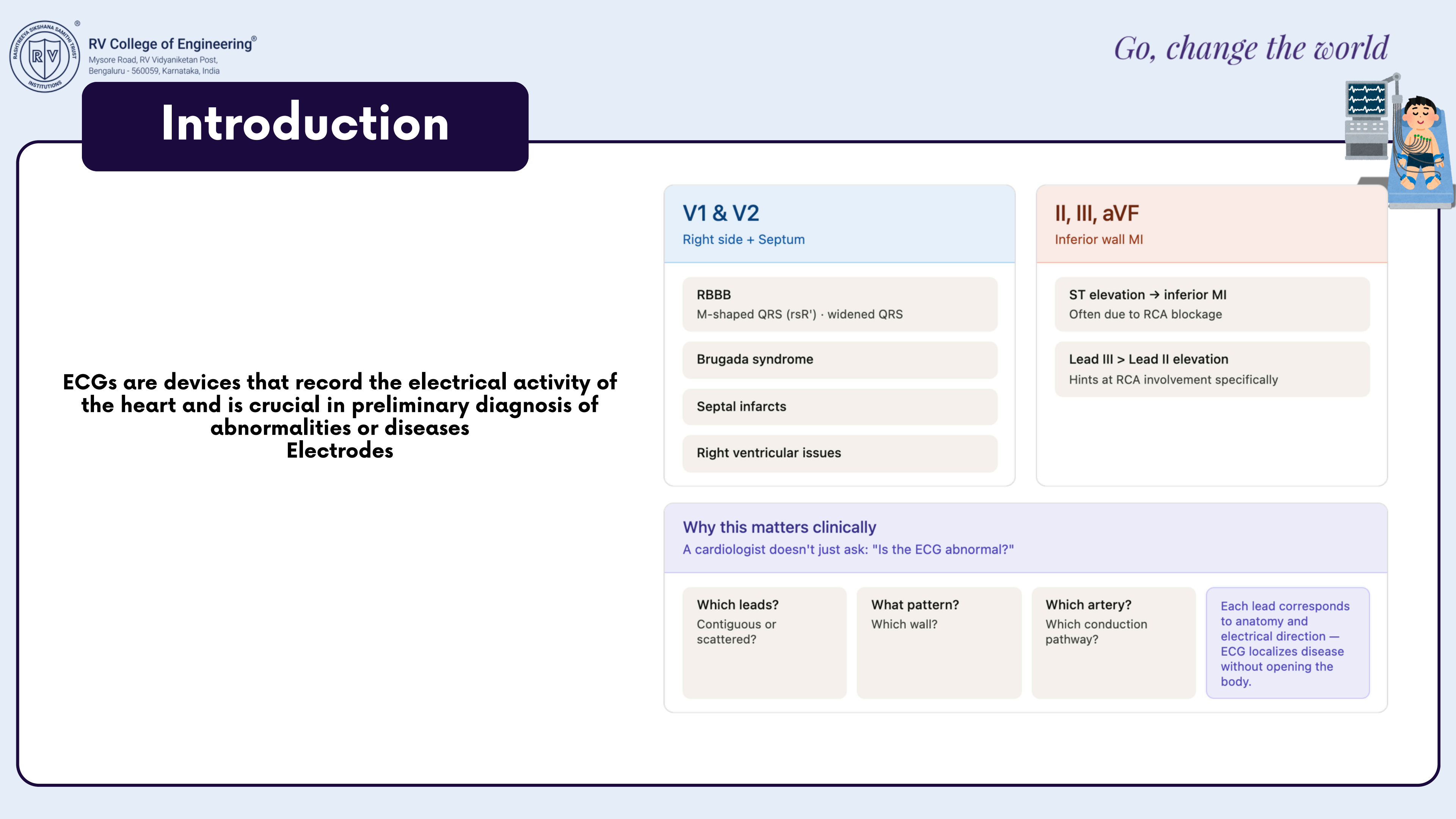


AUTOMATED AND ALIGNMENT- INVARIANT DIGITIZATION OF PAPER ECGS



Contents

- 1.Introduction
- 2.Existing system
- 3.Identification of problem
- 4.Literature Review
- 5.Problem definition
- 6.Objectives
- 7.Methodology
- 8.Tools and techniques.
- 9.Outcome.
- 10.Conclusion
11. References



Introduction



ECGs are devices that record the electrical activity of the heart and is crucial in preliminary diagnosis of abnormalities or diseases
Electrodes

V1 & V2
Right side + Septum

RBBB
M-shaped QRS (rsR') · widened QRS

Brugada syndrome

Septal infarcts

Right ventricular issues

II, III, aVF
Inferior wall MI

ST elevation → inferior MI
Often due to RCA blockage

Lead III > Lead II elevation
Hints at RCA involvement specifically

Why this matters clinically
A cardiologist doesn't just ask: "Is the ECG abnormal?"

Which leads?
Contiguous or scattered?

What pattern?
Which wall?

Which artery?
Which conduction pathway?

Each lead corresponds to anatomy and electrical direction — ECG localizes disease without opening the body.

Introduction



P wave — atrial activity

P wave duration < 120 ms

WHAT DOCTORS LOOK AT

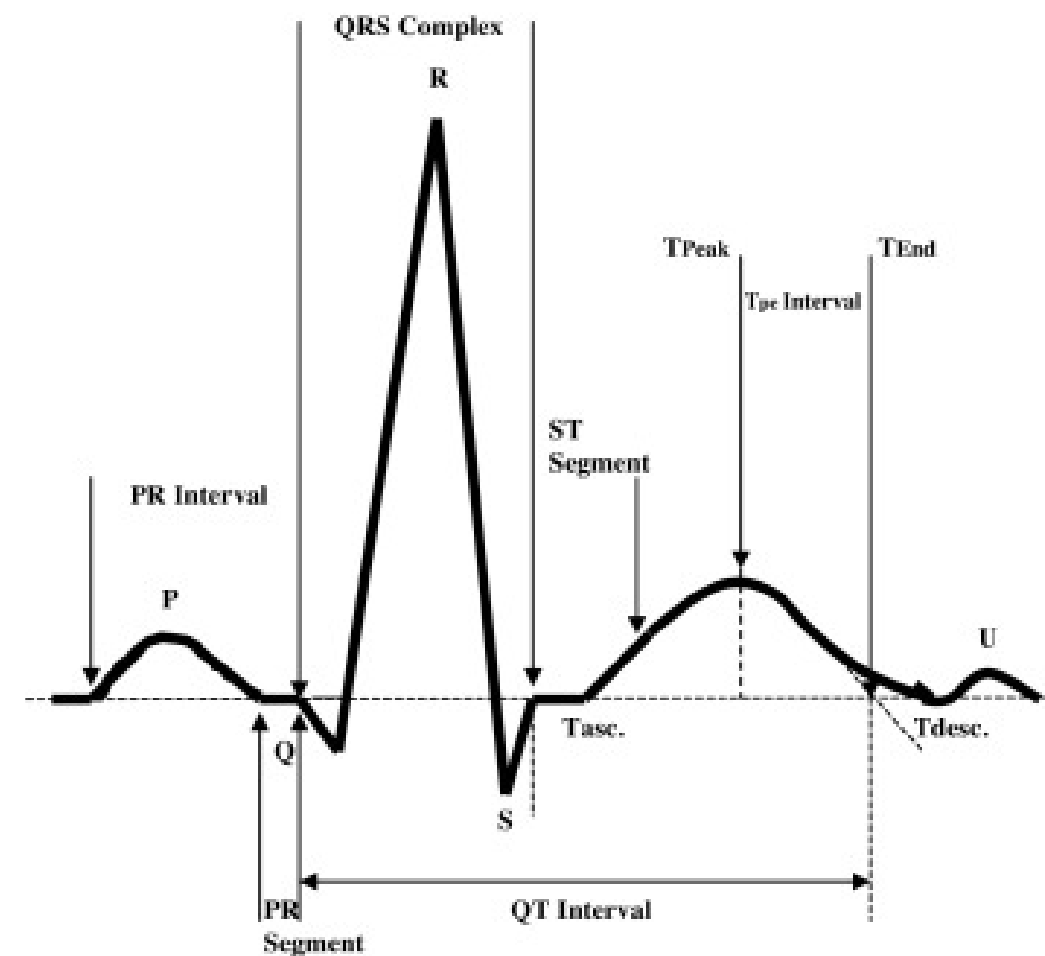
- Width
- Height
- Shape

EXAMPLES

- Tall peaked P waves → possible right atrial enlargement
- Broad/notched P waves → left atrial enlargement
- Missing P waves → atrial fibrillation

Why digitization accuracy matters:

If the signal is noisy or low-resolution, small P waves may disappear entirely. Bad filtering can smooth out notches or peaks that are diagnostically important.



PR interval — conduction delay

PR interval = 120 to 200 ms

EXAMPLES

- Long PR interval → first-degree heart block
- Gradually increasing PR → Mobitz I block
- Short PR interval → Wolff–Parkinson–White syndrome

Why accuracy matters:

ECG diagnosis often depends on differences of just a few milliseconds. Poor sampling rates can shift where the P wave or QRS starts/ends.

Example: At 25 mm/s ECG speed: 1 small box = 40 ms. A digitization error of even 1–2 pixels can affect diagnosis.

Introduction



QRS complex — ventricular depolarization

QRS duration < 120 ms

This is one of the most critical parts.

EXAMPLES

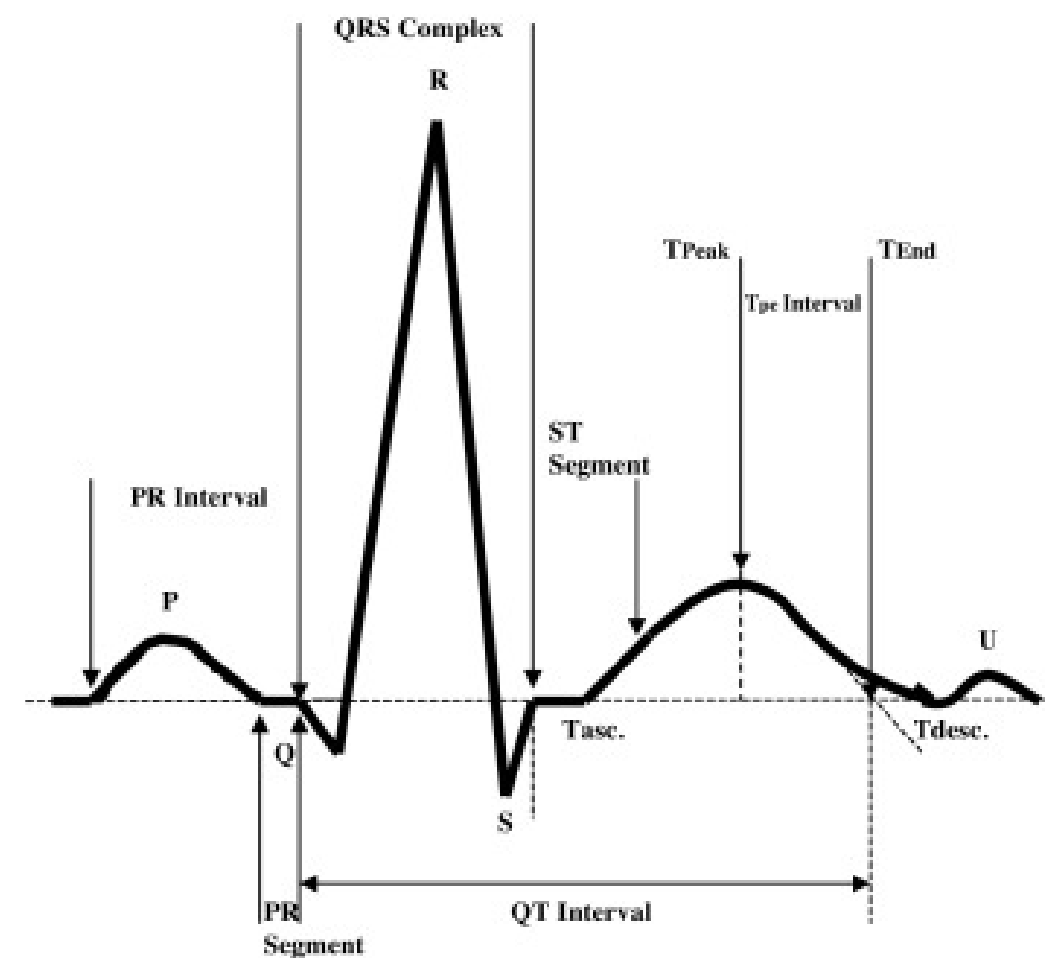
- Wide QRS → bundle branch block or ventricular rhythm
- Very tall QRS → ventricular hypertrophy
- Abnormal Q waves → previous myocardial infarction (heart attack)

Why accuracy matters:

Slight widening changes diagnosis. Oversmoothing can blunt sharp R peaks. Incorrect baseline correction can alter Q wave depth.

For automated systems: R-peak detection is foundational for: heart rate calculation, arrhythmia detection, HRV analysis.

Even small peak-detection errors propagate everywhere.



ST segment — extremely critical

ST elevation ≥ 1 mm

EXAMPLES

- ST elevation → possible acute heart attack (STEMI)
- ST depression → ischemia
- Saddle-shaped ST elevation → pericarditis

Why accuracy matters: This is huge in digitization.

Doctors may diagnose STEMI from tiny elevations: 1 mm can matter clinically. Baseline drift or poor calibration can fake ST elevation/depression. This is one of the biggest reasons ECG digitization systems need high fidelity.

Introduction



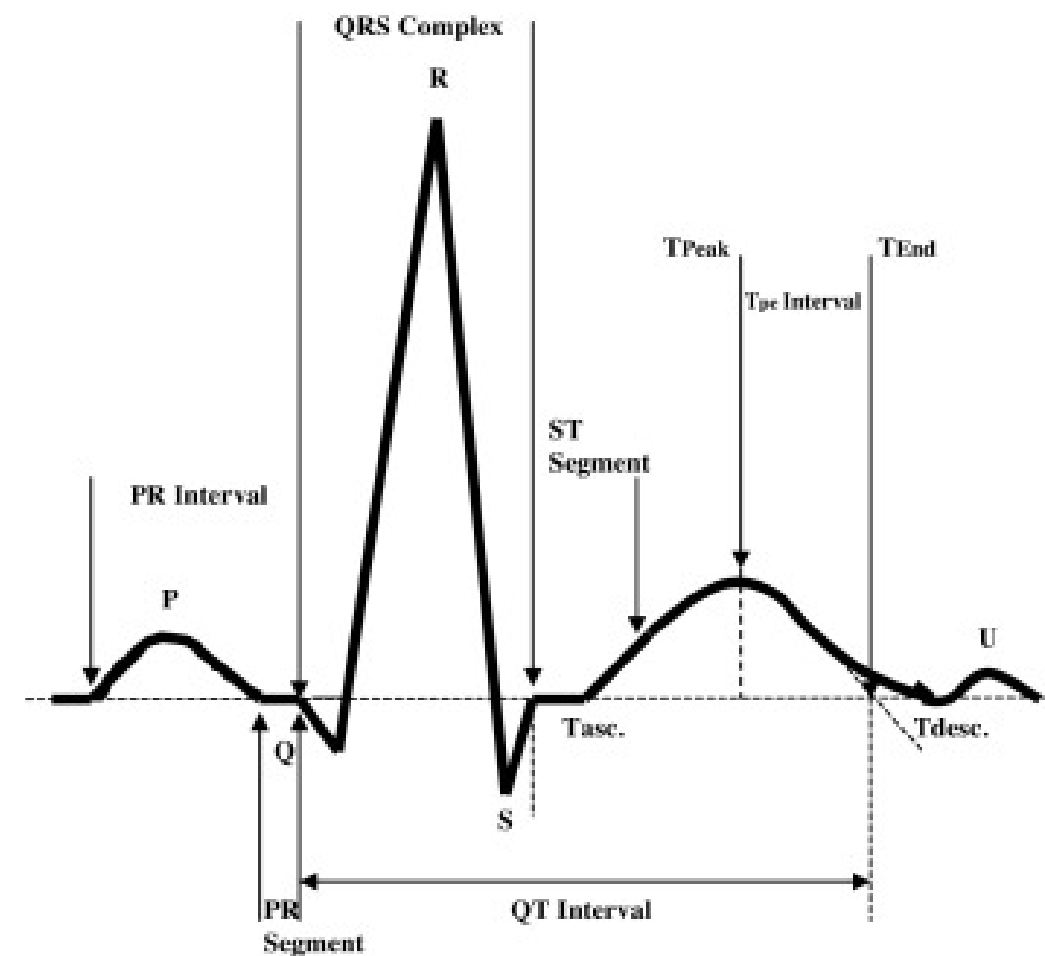
T wave — repolarization

EXAMPLES

- Tall peaked T waves → hyperkalemia
- Inverted T waves → ischemia
- Flattened T waves → electrolyte imbalance

Why accuracy matters:

T waves are smoother and lower frequency. Aggressive filtering can distort them badly. A digitized ECG might accidentally: flatten a T wave, create fake inversion, change amplitude ratios.



QT interval — risk of fatal arrhythmias

$$QTc = QT / \sqrt{RR}$$

EXAMPLES

- Long QT → risk of torsades de pointes (dangerous arrhythmia)
- Short QT → rare channelopathies

Why accuracy matters:

QT measurement depends on: exact T-wave ending, exact QRS start. Even experts disagree sometimes because the endpoint is subtle. Digitization errors here can affect: medication safety, arrhythmia risk prediction.

Introduction



Why this matters for ECG digitization specifically

If you're converting paper ECGs or analog ECGs into digital signals, common errors include:

COMMON ERRORS

- Gridline interference
- Skewed scans
- Baseline wander
- Low sampling resolution
- Peak clipping
- Noise filtering artifacts

THESE DIRECTLY ALTER

- Amplitudes
- Intervals
- Morphology

ECG diagnosis is heavily morphology-based.
So accuracy is not just "nice to have."

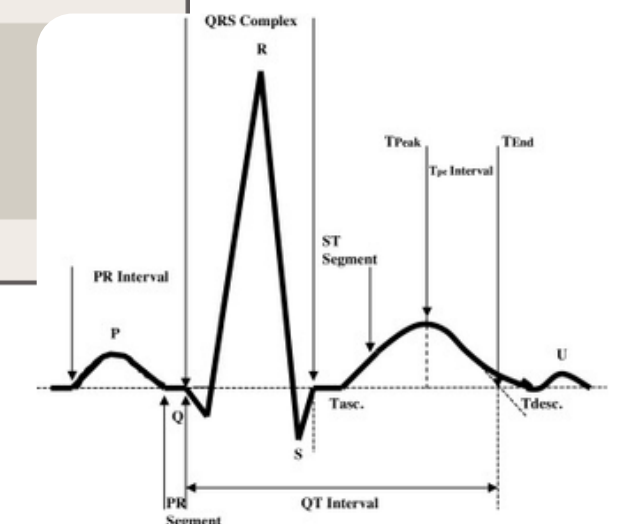
MEDICAL-GRADE STANDARDS REQUIRE

- Sampling frequency
- Amplitude resolution
- Filtering
- Calibration
- Signal preservation

"normal" → "heart attack"

"safe drug" → "dangerous QT
prolongation"

"sinus rhythm" → "atrial fibrillation"



Existing System

Automated Algorithm-Based Systems

- Detect anchor points & extract ECG signals automatically.
- Example: Wu et al. achieved 96% signal correlation using morphological algorithms and text-based lead recognition.
- Limitation: Sensitive to image quality, alignment, and printing variations.

Camera-Based / Point-of-Care Systems

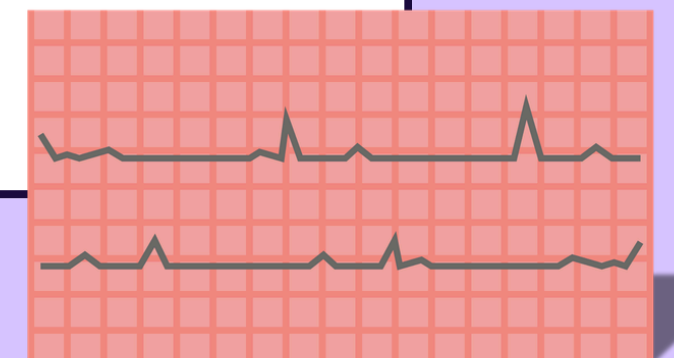
- Use cameras or smartphones to capture ECG images.
- Techniques: HSV color space analysis, adaptive illumination compensation.
- Limitation: Accuracy drops with non-uniform lighting or historical paper ECGs.

Traditional Image Processing Approaches

- Methods: Hough Transform (rotation), morphological filtering (QRS detection), adaptive binarization.
- Limitation: Require careful parameter tuning for different image types.

Deep Learning-Based Methods

- U-Net and nnU-Net architectures for ECG segmentation.
- Can handle complex backgrounds and lead separation.
- Limitation: Need large, diverse training data; may struggle with varying scan qualities.

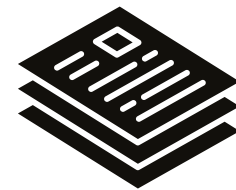


Identification of Problem

01

Large volumes of ECG data remain paper-based

especially in resource-limited healthcare settings, making them inaccessible for computational analysis and machine learning



02

Paper ECG records suffer from degradation

including ink fading, deterioration, scanning artifacts and loss of signal quality over time



03

Existing digitization methods are highly quality-dependent

performing reliably only on high-resolution, well-aligned, and well-illuminated ECG images



04

Traditional image processing techniques require manual tuning

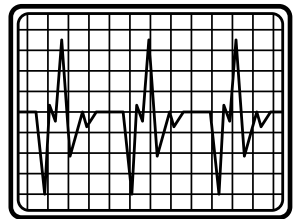
making them unsuitable for large, heterogeneous ECG archives



05

Real-world ECG scans exhibit significant variability

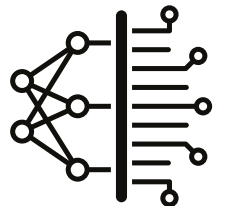
- Misalignment and rotation
- Inconsistent grid patterns and printing standards
- Noise from wrinkles, annotations, and poor lighting



06

Current deep learning approaches lack robustness and generalization

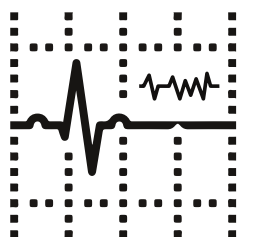
as they are often trained on limited or idealized datasets



07

Full-page ECG digitization is complex

requiring simultaneous lead localization, segmentation, grid removal, and accurate signal reconstruction





Literature Survey

Paper Title	All Authors (Full Names)	Published In (Year)	DOI / Source	Key Findings / Method	Results	Limitations
Combining Hough Transform and Deep Learning Approaches to Reconstruct ECG Signals From Printouts	Florian Krones, Ben Walker, Thomas Lyons, Ahmed Mahdi	arXiv (2024)	arXiv:2410.14185	Hough Transform for rotation correction + nnU-Net for segmentation	SNR = 17.02 dB on 21,799 ECGs	Sensitive to severe artifacts and scan degradation
A Fully-Automated Paper ECG Digitisation Algorithm Using Deep Learning	Hao Wu, Chen Xie, Jie Zhang, Lili Chu, Ming Li, Steve M. Plis	Scientific Reports (2022)	10.1038/s41598-022-25284-w	Anchor-point detection + morphological processing	96% signal correlation, MAE = 0.04 mV	Limited robustness to poor-quality scans
Paper-Recorded ECG Digitization Method with Automatic Restoration	Li-Hsiang Wang, Jaeho Kim, Miguel A. Rodriguez, Gari D. Clifford	Sensors (2024)	10.3390/s24175486	HSV color-space grid separation with illumination compensation	Real-time digitization capability	Performance affected by lighting variability
Fusion of Deep Learning and Rule-Based Techniques for Paper-Based ECG Digitization	Abdullah Kazi, Khanh Nguyen, Vijay Sendilraj, Sandeep Sinha, Madhusudhanan Natarajan	CinC (2024)	10.22489/CinC.2024	Rule-based preprocessing + deep learning segmentation	Improved accuracy over classical methods	Manual rule tuning required
nnU-Net: A Self-Configuring Method for Deep Learning-Based Biomedical Image Segmentation	Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, Klaus H. Maier-Hein	Nature Methods (2021)	10.1038/s41592-020-01008-z	Automated architecture configuration and preprocessing	State-of-the-art segmentation	High computational cost
Deep Learning-Based ECG Segmentation for Delineation of Diverse Arrhythmias	Changsoo Joung, Mircea Popescu, Konstantin Barsoukov, John C. S	PLOS ONE (2024)	10.1371/journal.pone.0303178	Multi-class ECG wave segmentation using U-Net	High delineation accuracy	Focused on clean ECG signals



Literature Survey

Paper Title	All Authors (Full Names)	Published In (Year)	DOI / Source	Key Findings / Method	Results	Limitations
From Paper to Digital: ECG Processing with U-Net Digitization and ResNet Classification	Xue Yu, Yuxin Huang, Jun Wu, Jun Wang, Weidong Cai	CinC (2024)	10.22489/CinC.2024.134	End-to-end ECG digitization and classification	F1 = 0.89	Limited evaluation on low-quality ECGs
Dual Deep Learning System to Digitize and Classify 12-lead ECGs from Scanned Images	Chia-Tung Chou, Santiago González	CinC (2024)	Semantic Scholar ID: 270969077	YOLOv7 + CNN + EfficientNet-B0	Improved classification accuracy	High model complexity
ECG-Image-Kit: A Synthetic Image Generation Toolbox	Kodthalu Shivashankara, Deepanshi, Mehri Shervedani, Mohammad A. Reyna, Gari D. Clifford, Reza Sameni	Physiological Measurement (2024)	10.1088/1361-6579/ad4954	Synthetic ECG image generation with realistic artifacts	Enables large-scale training	Synthetic data bias possible
PTB-XL: A Large Publicly Available Electrocardiography Dataset	Patrick Wagner, Natalia Strodthoff, Rolf T. Bousseljot, Dagmar Kreiseler, Andreas I. Lundin, Stefan E. W. Engquist, Carl-Johan Carlhäll, Markus Dillner, David V. Dyverfeldt, Anna Östling	Nature Scientific Data (2020)	10.1038/s41597-020-0495-6	Large-scale clinical ECG dataset	21,837 ECGs, 12 leads	No ECG images
PTB-Image: A Scanned Paper ECG Dataset	Xiaoyu Liu, Mohammad A. Reyna, Gari D. Clifford	arXiv (2024)	arXiv:2401.12345	Real scanned ECG images for digitization benchmarking	Mean SNR ≈ 0.01 dB	Limited dataset size
The George B. Moody PhysioNet Challenge 2024	Mohammad A. Reyna, Deepanshi, Jacob Weigle, Zuzana Koscova, Kyle Campbell, Kodthalu Shivashankara, Shayan Saghafi, Sara Nikookar, Mahdi Motie-Shirazi, Yasaman Kiarashi, Saeed Seyedi, Maryam Hassannia, Anne M. Bjørnstad, Erik Stenhede, Amir Ranjbar, Gari D. Clifford, Reza Sameni	CinC (2024)	10.22489/CinC.2024	Standardized ECG digitization evaluation	Introduced ASC & SNR metrics	Challenge-specific constraints

Problem Definition

- A large number of ECG records are still available only in paper format, making them unsuitable for automated analysis and long-term digital storage.
- Full-page ECG digitization is complex, as it requires simultaneous lead localization, segmentation and signal extraction, making it highly sensitive to noise, misalignment, and layout variations in real-world scans.
- Existing approaches treat the ECG as a single entity, leading to reduced accuracy and poor generalization under diverse conditions.
- Lack of deep learning frameworks that decompose ECGs into individual leads for simpler and more reliable learning.
- **Design a lead-wise deep learning approach that isolates and segments each lead independently to improve accuracy and scalability in real-world ECG digitization.**

Why nnUnet?

nnU-Net was chosen because it is a complete biomedical segmentation framework, not just a model, and it automatically handles preprocessing, training setup, patch size selection, and useful data augmentations for the dataset.

Objectives

Digitize paper-based ECG records into accurate and clinically usable one-dimensional digital signals suitable for automated analysis and long-term electronic storage.

1

Develop a processing pipeline capable of handling skewed, rotated, and layout-diverse ECG scans without requiring manual alignment or correction.

2

Perform ECG signal trace segmentation using an nnU-Net-based model trained on synthetically generated image-mask pairs, eliminating the need for manual pixel-level annotation.

3

Use yolov8 or an equivalent lightweight model to crop out each individual lead, ensuring that the system is lead independent and the complexity for the model is lesser

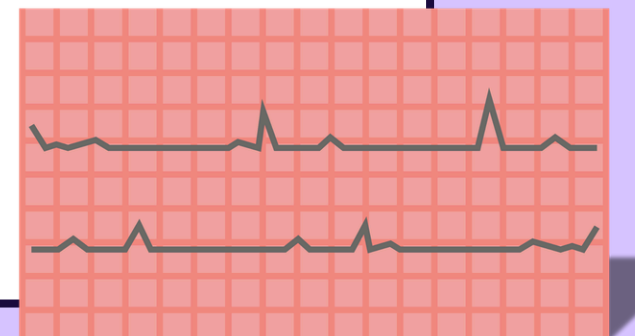
4

Convert segmented ECG traces into continuous and clinically calibrated digital waveforms expressed in standard time (milliseconds) and amplitude (millivolts) units.

5

Evaluate system performance using established segmentation metrics such as Dice Similarity Coefficient and Intersection over Union on large-scale benchmark datasets.

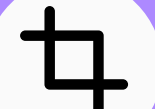
6



Methodology



Stage 1
ECG Image
Generation



Stage 2
Lead
Cropping



Stage 3
Signal
Trace
Segmentation



Stage 4
Vectorization
and
Calibration

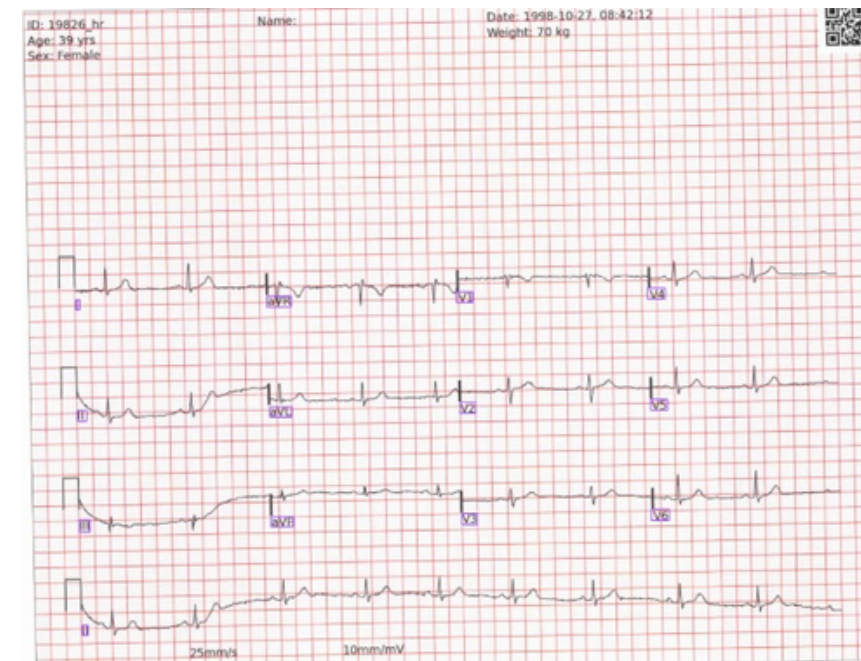
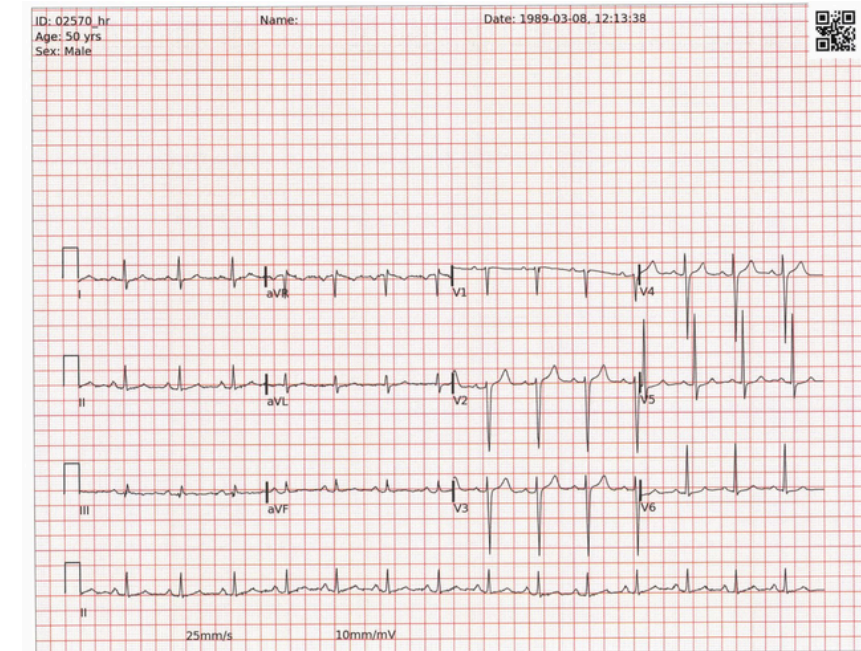
Methodology

1. ECG Image Generation

- ECG signals from PTB-XL (.hea + .dat) are converted into synthetic paper-style ECG images generated by ecg-image-kit library
- These images replicate paper grids, noise, distortions, and layout variations.
- This forms the input to the entire pipeline.

2. Lead Cropping

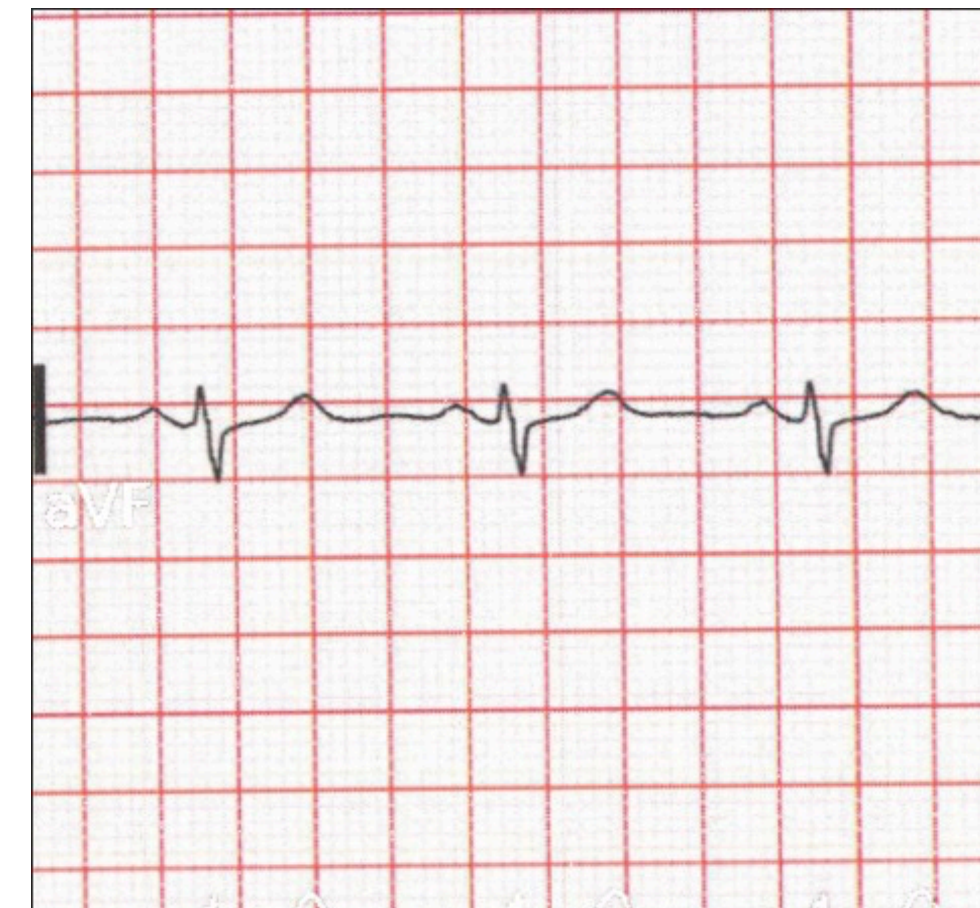
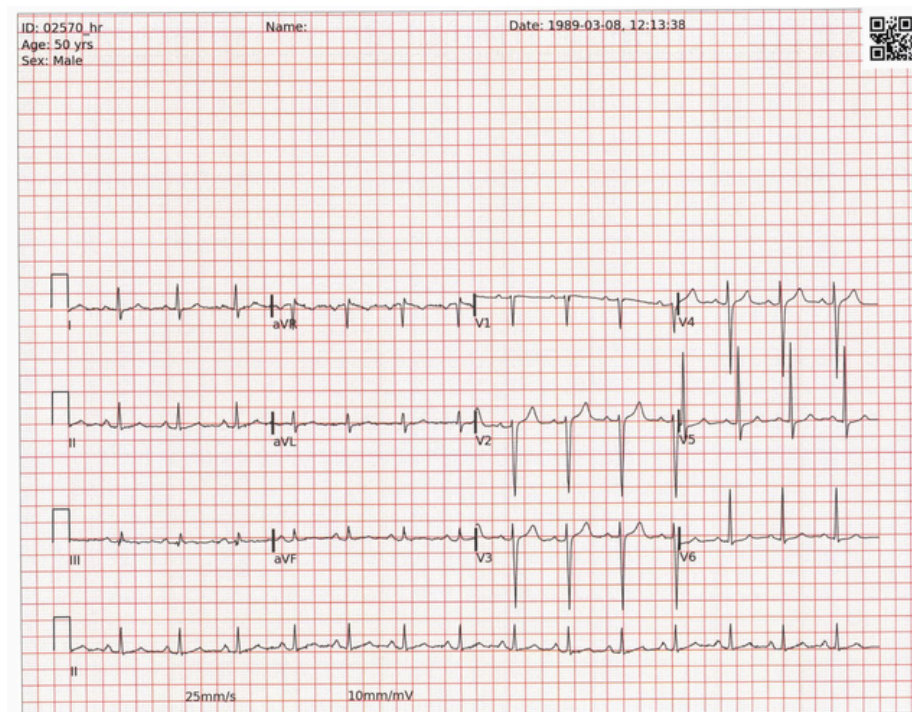
- Use YOLOv8 (think) or an equivalent lightweight object detection model to accurately crop out each individual ECG lead from the input image.
- Ensure the system remains lead-independent, thereby reducing model complexity and improving efficiency by processing each lead separately.



Methodology

2. Lead Cropping

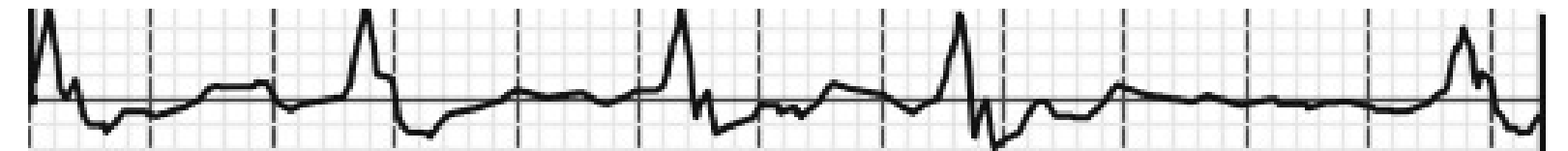
- This approach adapts automatically to different ECG layouts (3×4 , 2×6) and corrects minor rotation.
- Output: 12 clean, standardized lead images per ECG.



Methodology

3.Signal Trace Segmentation

- Each cropped lead image is treated as a binary segmentation problem.
- The goal is to separate ECG signal traces from grid lines and background.
- Segmentation is performed using the nnU-Net framework.
- Training data is synthetically generated using ECG-Image-Kit, based on PTB-XL time-series signals.
- Pixel-level ground truth masks are generated automatically, avoiding manual annotation.
- The model is trained using Soft Dice Loss to handle class imbalance.

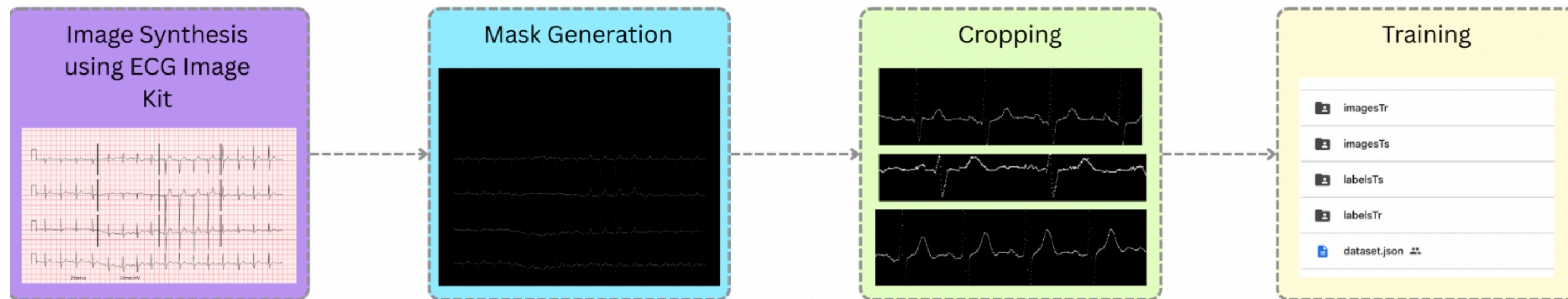
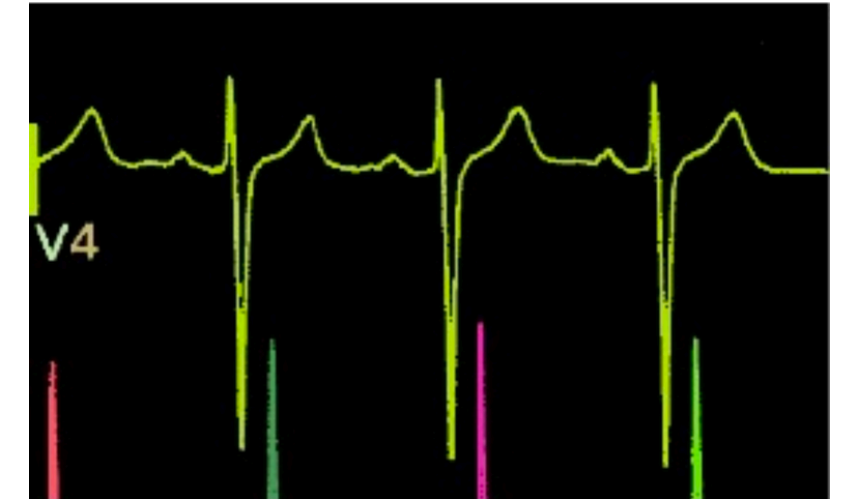


$$\text{Dice}(p, g) = \frac{2 \sum_i p_i g_i}{\sum_i p_i + \sum_i g_i}$$

Methodology

3.Signal Trace Segmentation

- The model is trained using Soft Dice Loss to handle class imbalance.
- 5-fold cross-validation is used, and predictions are obtained via ensemble averaging.
- The cropped lead images + corresponding masks are fed into nnU-Net.
- Connected component analysis is applied to each crop, retaining only the largest component to remove noise and interference from adjacent leads.
- Output: binary masks isolating ECG signal pixels for each lead.



Methodology

4.Vectorization and Calibration

Vectorization

- The binary segmentation mask is converted into a 1D signal.
- For each time column, the median y-coordinate of signal pixels is selected.
- This produces a single-pixel-wide centerline representing the ECG waveform.

Calibration

- Pixel coordinates are converted into physical units.
- ECG grid spacing is used to compute pixel-to-millimeter scaling factors.

$$\text{Time (ms)} = \frac{(p_x - p_{x0})}{S_x} \times 40$$

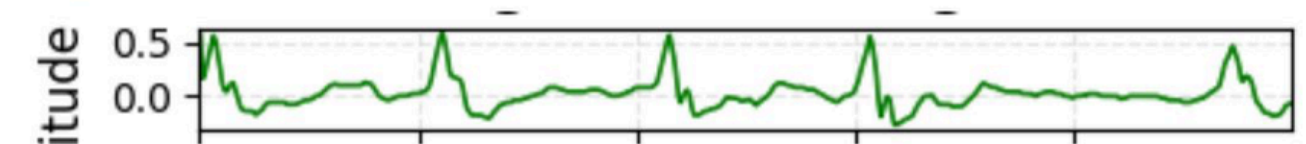
$$\text{Amplitude (mV)} = \frac{(p_{y0} - p_y)}{S_y} \times 0.1$$

- Output: clinically calibrated ECG signals in milliseconds and millivolts.

Segmentation Mask



Signal Plot



ecg_signal	
Time(s)	Amplitude(mV)
0.0000	1.21999990940094
0.0020	1.1093919277191200
0.0040	0.6223199367523190
0.0060	0.13524791598320000
0.0080	-0.011972030624747300
0.0100	-0.00018801917030941700
0.0120	0.01159598957747220
0.0140	0.0233799796551466
0.0160	0.035163965076208100
0.0180	0.039999961853027300
0.0200	0.039999961853027300
0.0220	0.039655961096286800
0.0240	0.03179997205734250
0.0260	0.02394397370517250
0.0280	0.019999980926513700
0.0300	0.019999980926513700

Methodology

Grid Detection

- Accurate ECG grid detection is challenging because the waveform itself overlaps and obscures the underlying grid lines. Direct grid extraction often fails in regions where the ECG trace intersects the grid, leading to inaccurate pixel-to-physical unit calibration.
- To address this issue, the segmentation mask generated by the nnU-Net model is utilized to identify ECG signal pixels. These signal pixels are then removed through a local background reconstruction process, where each signal pixel is replaced using the intensity characteristics of neighboring non-signal pixels. This effectively restores the appearance of the original ECG paper background while preserving the grid structure.

Step 1: Signal Isolation

- The nnU-Net model produces a binary mask that accurately separates ECG waveform pixels from the background.

Step 2: Local Background Reconstruction

- Pixels belonging to the ECG waveform are replaced using information from surrounding background pixels. This suppresses the signal trace and reconstructs a continuous grid pattern, making the major grid lines significantly more prominent.

Step 3: Major Grid Detection

- After reconstruction, the dominant 5 mm grid lines are detected by analyzing vertical and horizontal intensity profiles. The detected major grid positions are:

[38, 79, 123, 158, 196, 236, 276, 306, 353, 394, 433]

The average spacing between adjacent major grid lines is:

Major grid spacing = 40 pixels

Methodology

Step 4: Pixel-to-Physical Unit Calibration

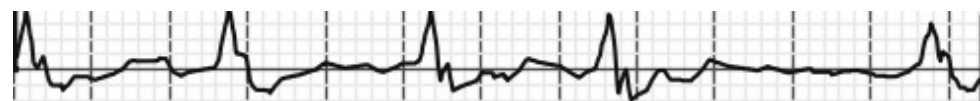
Since a major ECG grid square corresponds to 5 mm, the physical scale is computed as:

Thus:

- Spatial Resolution: 0.125 mm/pixel
- Temporal Resolution: 0.005 s/pixel (assuming standard ECG speed of 25 mm/s)
- Amplitude Resolution: 0.0125 mV/pixel (assuming standard ECG gain of 10 mm/mV)

Calibration Parameters Used for Signal Reconstruction

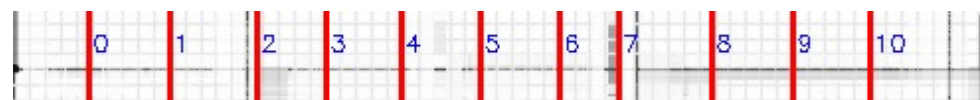
For waveform digitization and subsequent ECG analysis, the primary calibration factors employed are 0.125 mm/pixel and 0.005 s/pixel, which establish the spatial and temporal scaling required to convert image coordinates into clinically meaningful measurements.



Clean ECG Image with waveform obscuring grid



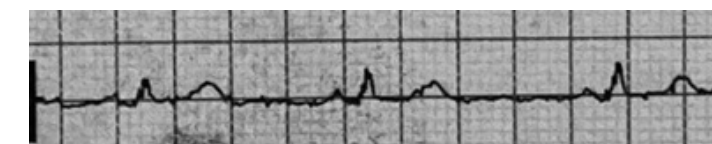
After signal removal / background reconstruction



Gridlines detected

$$\text{mm per pixel} = \frac{5}{40} = 0.125 \text{ mm/pixel}$$

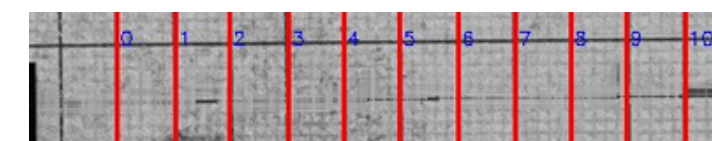
Parameter	Value
Major Grid Spacing	40 px
mm per Pixel	0.125 mm/px
sec per Pixel	0.005 s/px
mV per Pixel	0.0125 mV/px



Degraded ECG Image with waveform obscuring grid

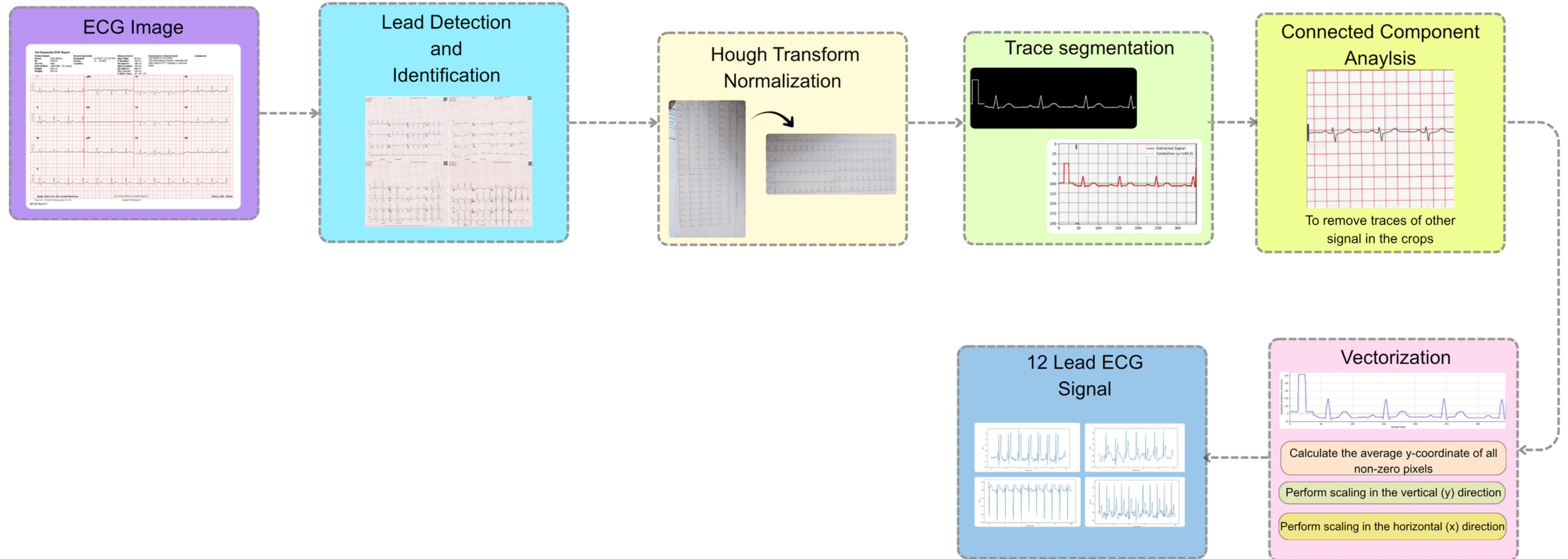


After signal removal / background reconstruction



Gridlines detected

Tools and techniques



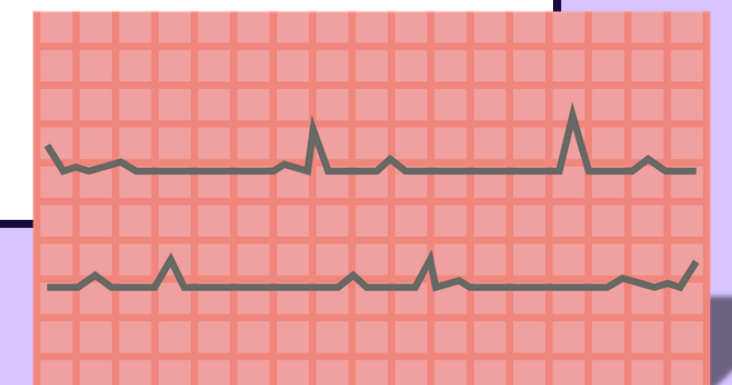
Outcomes

Qualitative Results

- The segmentation model accurately isolated ECG signal traces from real-world scanned paper ECGs.
- Successful performance was observed on both clean, high-quality scans and degraded scans with uneven lighting and paper artifacts.
- Visual overlays showed strong alignment between predicted masks and actual ECG waveforms, confirming effective signal separation from background grids and noise.



Segmentation results (blue mask) superimposed on original scans from the ECG-Image-Database. (a) A high-quality scan of a clean paper ECG showing precise signal isolation. (b) A scan of a degraded ECG paper captured under variable lighting conditions, demonstrating the model's robustness to environmental artifacts.



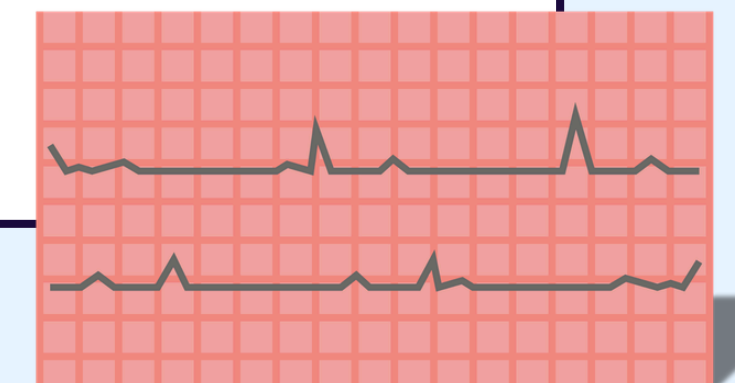
Outcomes

Quantitative Segmentation Performance

- Mean Dice Similarity Coefficient (DSC):
0.9246
- Mean Intersection over Union (IoU):
0.8609
- High overlap between predicted segmentation masks and ground truth annotations across validation folds.
- Lead-wise cropping prior to segmentation significantly reduced background complexity and improved segmentation accuracy.

Metric	Value	Description
Dice Similarity Coefficient (DSC)	0.9246	Harmonic mean of precision and recall; the primary measure of segmentation accuracy.
Intersection over Union (IoU)	0.8609	Quantifies the area of overlap between the predicted segmentation and the ground truth.
Precision	0.9315	The ratio of correctly predicted signal pixels to total predicted signal pixels.
Recall (Sensitivity)	0.9122	The ratio of correctly predicted signal pixels to actual signal pixels in the ground truth.

$$\text{Dice}(p, g) = \frac{2 \sum_i p_i g_i}{\sum_i p_i + \sum_i g_i} \quad \mathcal{L}_{\text{Dice}} = 1 - \text{Dice}(p, g)$$

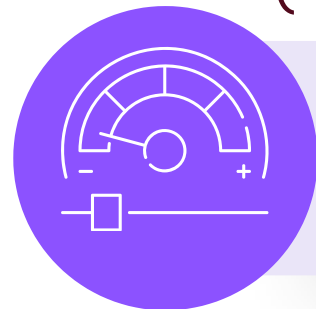


Outcomes



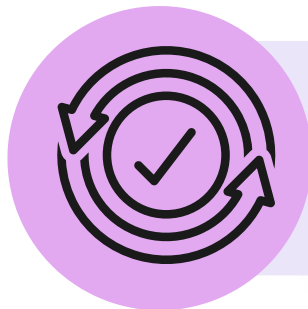
**Highest segmentation performance
observed on Lead aVR**

(Dice: 0.9667, IoU: 0.9355)



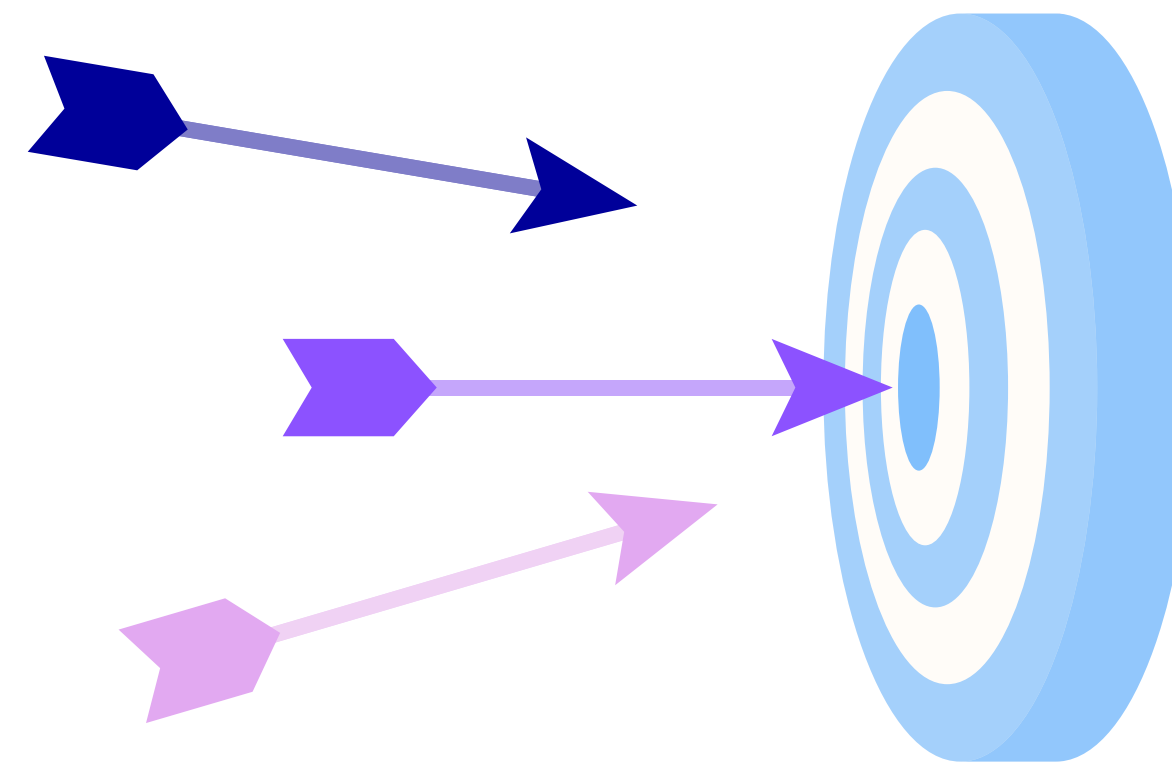
**Lowest performance observed
on Lead V4**

(Dice: 0.7853, IoU: 0.6465)

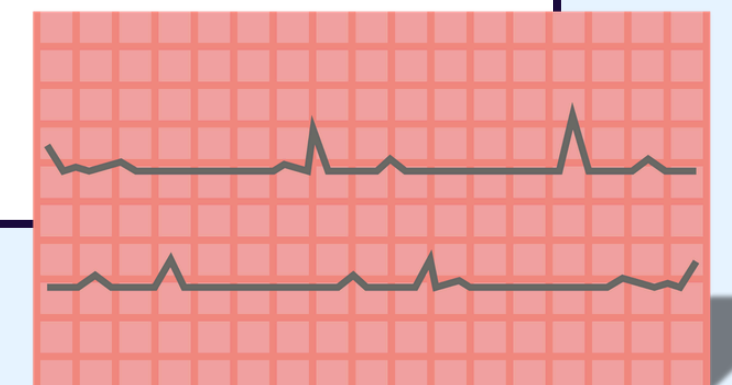


**Standard limb leads showed
consistent and reliable performance**

- Lead I Dice: 0.9289
- Lead II Dice: 0.9446

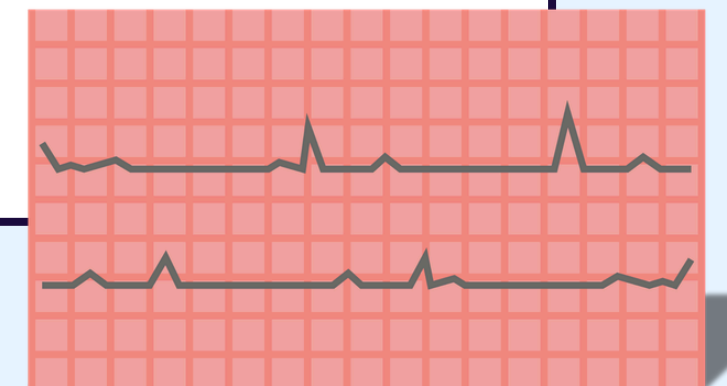
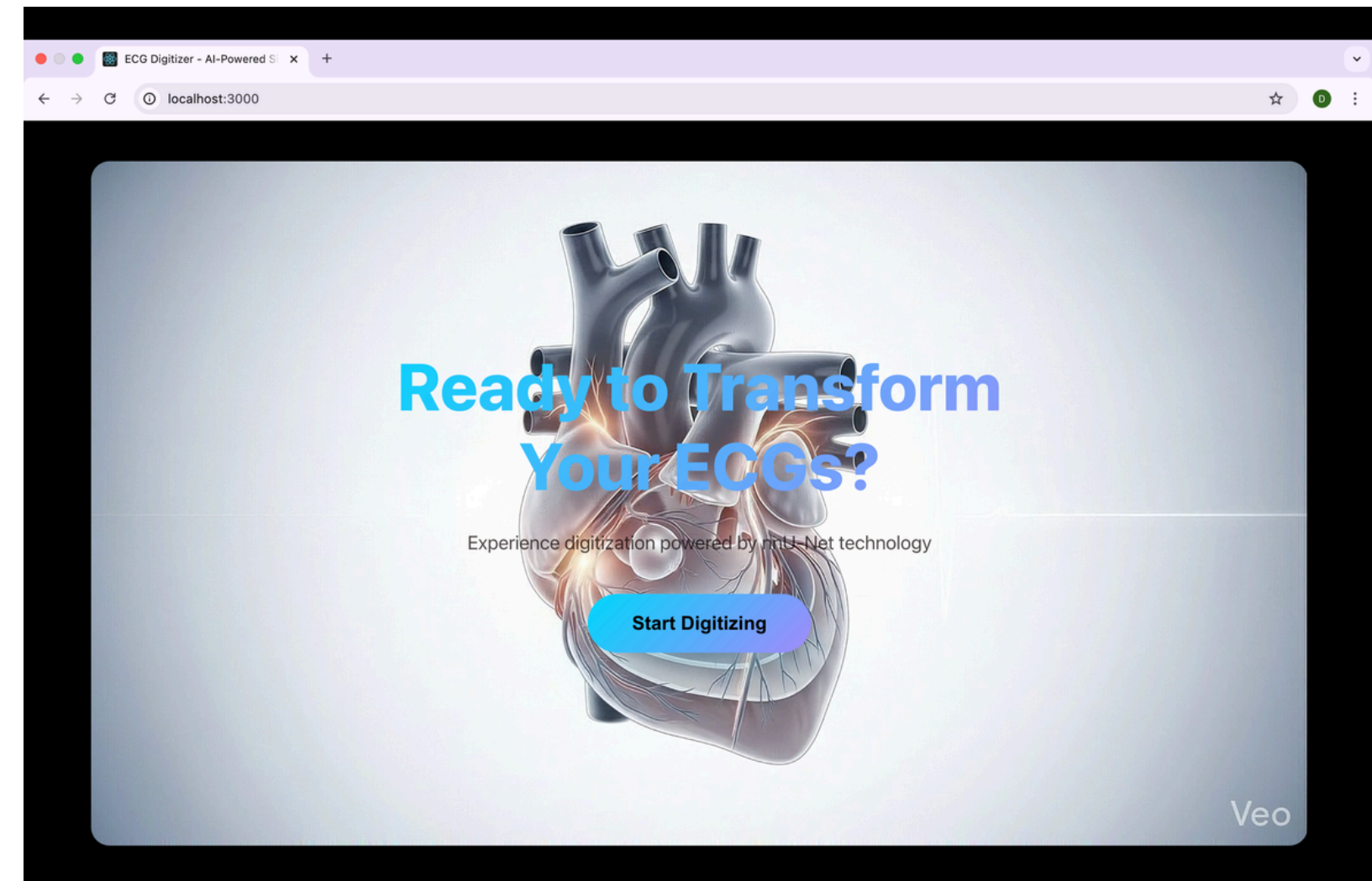
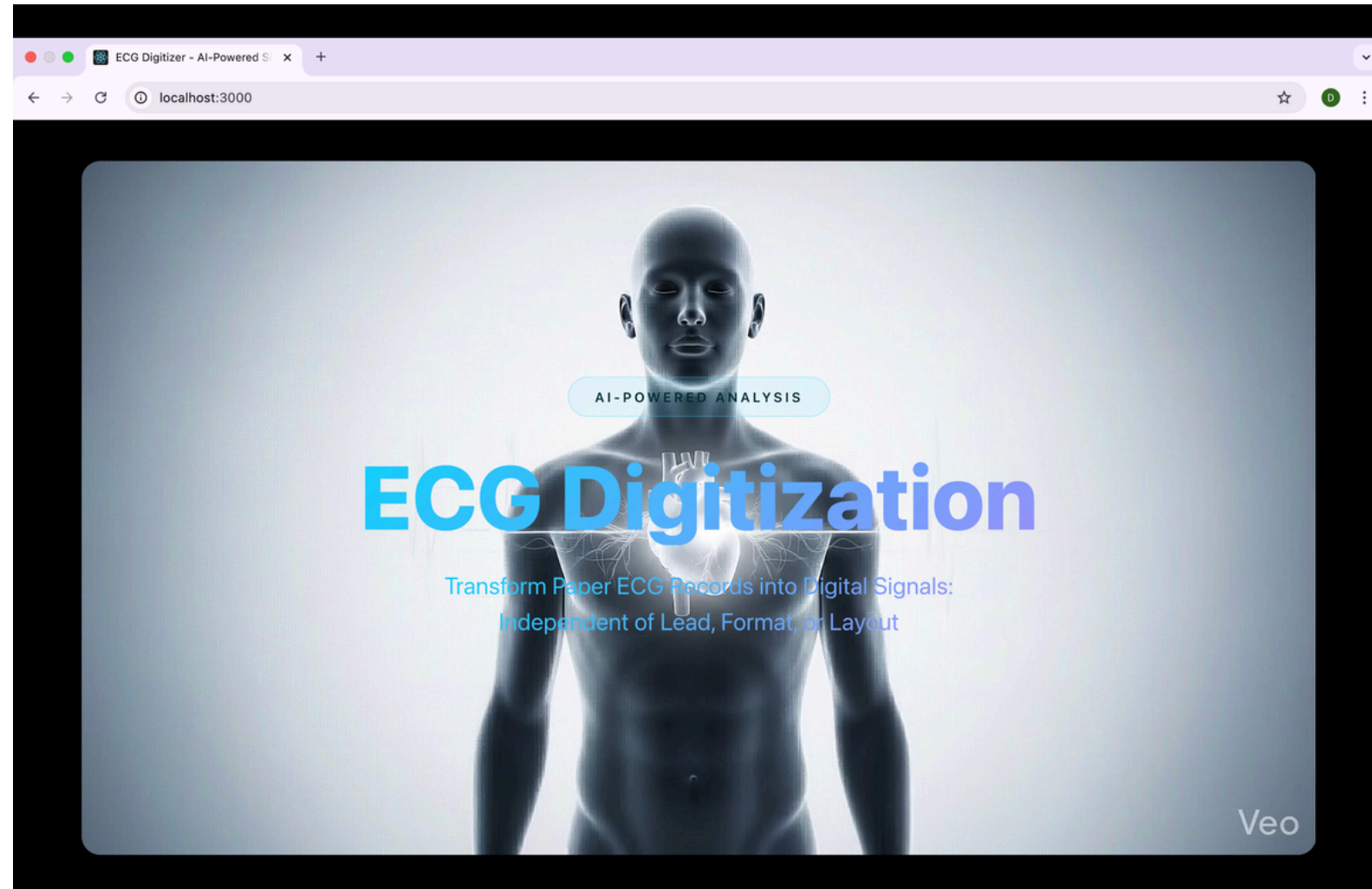


**Lead-Wise Performance
Variability**



Outcomes

Go, change the world





Outcomes

ECG Digitizer

AI-Powered Signal Extraction

Upload ECG Image

Drop your ECG scan or click to browse

Drop ECG image here

PNG, JPG, JPEG

Process ECG

No file chosen

Lead Information

Lead Name

V5, II, aVL...

Calibration

Seconds per pixel

0.004

Default: 0.004

mV per pixel

0.01

Default: 0.01

Y-shift ratio

0.5

Default: 0.5

ECG Digitizer

AI-Powered Signal Extraction

New Analysis

Analysis Complete

Signal extracted successfully

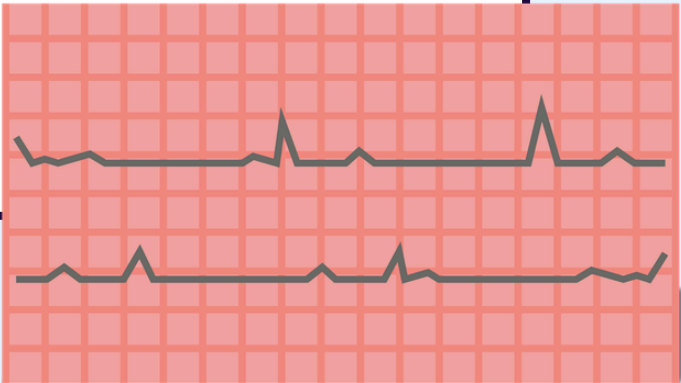
Signal Metadata

Sampling Rate	Duration	Samples	Precision
500 Hz	2.50s	1250	0.0100 mV

Segmentation Mask

Signal Waveform

Download CSV Data



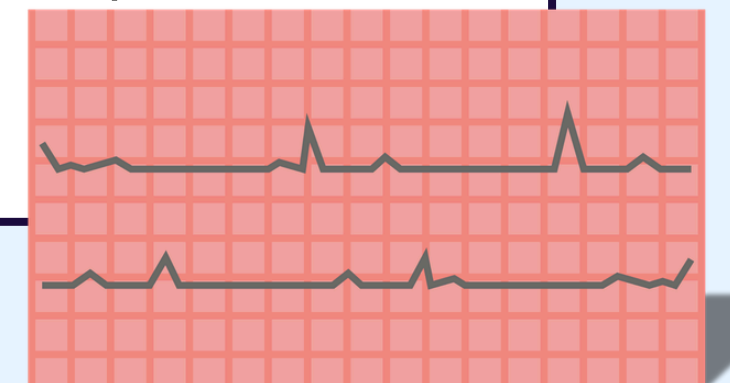
Conclusion

Financial Viability

- Low Maintenance: The use of nnU-Net, a self-configuring framework, automatically handles hyperparameters. This reduces the technical debt and the need for expensive Machine Learning Engineers to manually tune the model for different datasets.
- Architecture Modularity: The methodology is broken into four independent stages. This modularity allows for low-cost updates; for example, you can improve the "Vectorization" phase without needing to retrain the entire "Segmentation" model, keeping maintenance costs low.

ROI

- Unlocks legacy ECG archives by digitizing faded, degraded, and low-quality scanned records that would otherwise remain unusable.
- Reduces infrastructure cost by working with variable-quality scans from standard scanners/mobile devices, avoiding premium hardware dependence.
- Improves hospital efficiency by giving faster access to past ECG records, reducing repeat tests and supporting better patient history review.



Conclusion

DEPLOYMENT COST — THE CLINICAL PHASE

Inference cost

₹40-150

Per scan, running the AI model in a secure, compliant cloud environment.

Hardware

₹0 extra

Works with existing office
scanners or smartphones — no
specialized ECG hardware
needed.

Integration

₹1k-5k/mo

Standard AI-enabled EMR subscription in India to host the tool.

LATENCY-COST TRADEOFF

CPU-only (low cost)

2-5 min · non-viable

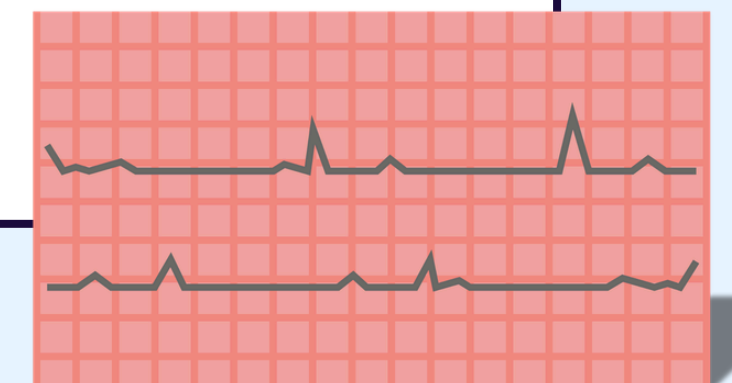
Mid-tier GPU **selected**

<15 sec · real-time ready

High-end GPU (max cost)

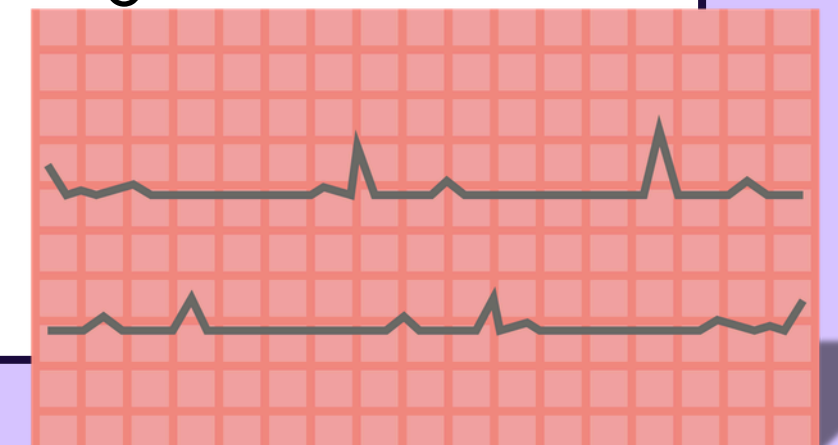
<5 sec · cost-prohibitive

To ensure the solution remains clinically desirable, we prioritized a mid-tier GPU configuration to resolve the latency-cost tradeoff, delivering a <15 second turnaround time necessary for real-time patient consultations rather than the non-viable, multi-minute delays associated with lower-cost, CPU-only environments.



Conclusion

- This work presents an effective method for digitizing paper-based ECGs using a deep learning-based segmentation approach.
- A synthetic data generation strategy was used to convert 1D ECG signals into 2D image-mask pairs, eliminating the need for manual annotation.
- By decomposing full-page ECG images into individual lead regions, the complex segmentation task was simplified into independent binary segmentation problems.
- Training the nnU-Net model on these lead-specific crops resulted in accurate pixel-level separation of ECG signals from background grids and noise.
- Quantitative evaluation using 5-fold cross-validation on the PTB-XL dataset demonstrated high segmentation accuracy, confirming the effectiveness of the proposed pipeline.
- This shows that lead-wise segmentation combined with synthetic training data is a practical and scalable approach for converting paper ECG archives into usable digital signals.



References

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